Federated Full-Envelope LPV Identification from Incomplete Local Coverage

***Abstract*—Vehicle identification data may cover only narrow speed regions, leaving full-envelope linear-parameter-varying (LPV) models undetermined locally. Fleet collaboration can supply missing scheduling information, but pooling vehicles with different dynamics introduces bias. We address this tradeoff through federated group regression. Clients estimate fixed-speed models from measured trajectories and known applied steering by output-error (OE) fitting. Additive statistics and local assignment scores then support learning shared LPV maps without exchanging trajectories or requiring complete local maps. Across ten independent simulated nonlinear vehicle fleets with family labels withheld, mean matrix error at locally unobserved speeds is 2.05%, compared with 3.94% for Smooth Local and 4.48% for Global, with positive adjusted paired confidence intervals for both improvements. Diagnostics show that accurate local estimates alone do not ensure sufficient group coverage or reliable extrapolation. Supporting linear-quadratic-integral (LQI) tests reuse the frozen maps and satisfy the tested constraints in all federated rollouts under noise-free state feedback. Tracking improves consistently over the fixed nominal controller, while gains over Global are modest and recipient-dependent.**

***Index Terms*—federated identification, LPV systems, incomplete operating coverage, heterogeneous vehicles**

I. Introduction

Model-based vehicle control requires dynamics models that remain useful as operating conditions change. For lateral motion, longitudinal speed affects the relationship between steering, sideslip, and yaw rate. A model identified near one operating point may therefore be inadequate elsewhere. LPV models represent this dependence explicitly, making identification across the intended speed range a central requirement [1], [2].

Obtaining that coverage separately for every vehicle is a demanding calibration task. A vehicle may provide measurements over only a narrow part of its intended operating range. Those measurements constrain the observed response, but a flexible speed-dependent model can remain undetermined outside that region. Repeating measurements at the same speeds reduces noise without resolving missing scheduling directions. The resulting problem is therefore incomplete operating information as well as limited sample count.

Fleet collaboration offers a way to fill these gaps: one vehicle's unobserved speeds may be covered by others. Yet differences in mass, inertia, and tire properties make their measurements only approximately compatible. Pooling all vehicles can introduce bias, while separating them too aggressively can leave each group without sufficient speed coverage. What is needed is selective sharing that supplies missing operating information without overwhelming a recipient's dynamics with incompatible data. Crucially, the partners must be chosen from partial observations, before complete local models are available.

We address this need through federated LPV identification. Vehicles keep trajectories locally, exchange additive statistics, and evaluate candidate group maps at their observed speeds. Alternating assignment and regression learns shared maps over the wider interval. The central question is whether groups inferred from partial data can provide both complementary coverage and a useful approximation of their recipients' dynamics. Figure 1 summarizes the procedure.

Existing work supplies important components of this approach. Local LPV identification constructs scheduling-dependent maps from operating-point models [3]. Federated learning provides a framework for collaboration without centralizing records [4], and FedSysID characterizes the effect of dynamical heterogeneity on collaborative identification [5]. IFCA alternates cluster assignment and model optimization [6]. Clustered SysID applies this structure to LTI systems with identical dynamics within each true cluster [7]. IC-SYSID adds incremental clustering and stability promotion [8], while [9] uses uncertainty-aware grouping for vehicle controller design.

Our distinction is the joint requirement of compatibility and complementary scheduling information. Each client's fixed speed support leaves its LPV regression deficient, and shared maps remain approximate within groups. The cited recovery guarantees do not establish recovery in this setting. We specialize the established alternating clustering structure to this problem and make the following contributions:

1) We formulate LPV identification from incomplete local speed coverage in common physical-state coordinates. A conditional rank statement distinguishes information supplied by pooling from extrapolation imposed by a local prior.
2) We develop a label-free federated procedure that learns shared LPV maps from local OE estimates and partial speed support. Additive statistics and local assignment scores allow grouping without complete local LPV maps.
3) We evaluate selective sharing against local priors and global pooling on ten independent vehicle fleets. Coverage and recipient diagnostics explain the scope of the identification gains, while LQI tests with the frozen maps distinguish aggregate model accuracy from recipient-level control performance.

Sections II and III present the identification problem and federated regression. Section IV describes the benchmark. Sections V and VI report identification and LQI tracking results. Section VII discusses the findings and concludes.

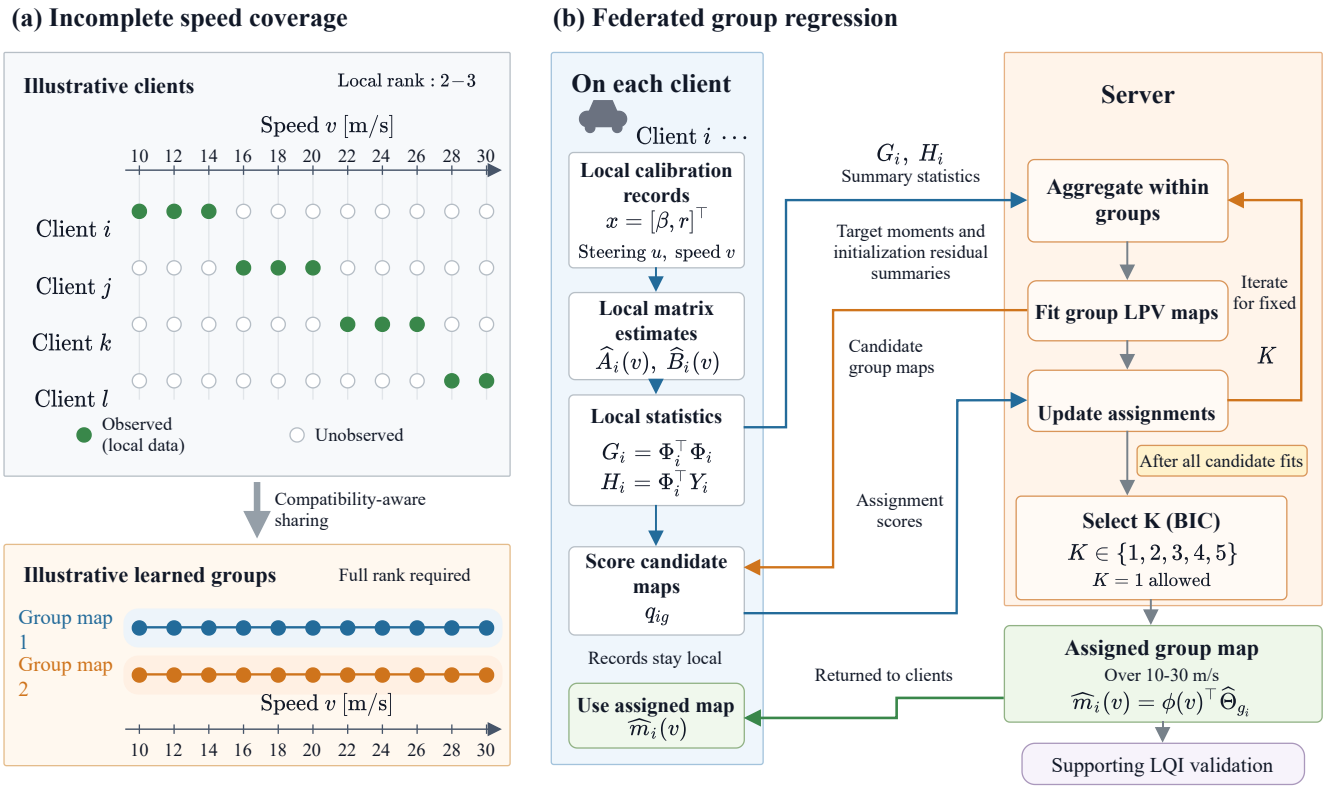


Fig. 1. Compatible federated LPV identification. (a) Illustrative clients and groups. Local speed blocks give scheduling rank two or three for a seven-function basis; sharing can supply missing directions when the pooled group design has rank seven. (b) Clients retain records and matrix targets, exchange additive statistics, residual summaries, and assignment scores, and receive group maps. Fitting and assignment alternate for each candidate K, followed by selection using the Bayesian information criterion (BIC). Local output-error fits use measured state trajectories and known applied steering at fixed speeds. The envelope is 10–30 m/s along the prescribed cornering curve; LQI checks downstream usability.

II. Identification Problem

Identification has two stages: estimating dynamics from local trajectories at observed speeds and constructing an LPV map over the intended interval. The second stage requires scheduling information that an individual vehicle may lack, motivating fleet sharing.

A. Local identification from measured trajectories

Consider M vehicles indexed by $i \in \{1, \dots, M\}$ and the target speed interval $\mathcal{V} = [v_{\min}, v_{\max}]$, with $0 < v_{\min} < v_{\max}$. Vehicle i has state $x_i = [\beta_i, r_i]^\top \in \mathbb{R}^2$, comprising sideslip angle β_i and yaw rate r_i, and front-wheel steering input $u_i \in \mathbb{R}$. Its measurements are $y_{i,k} = x_{i,k} + \nu_{i,k}$, where $\nu_{i,k} \in \mathbb{R}^2$ is state measurement noise and k indexes samples. Applied steering and longitudinal speed are assumed known.

Vehicle i supplies fixed-speed trajectories at n_i observed speeds $\mathcal{V}_i = \{v_{i,1}, \dots, v_{i,n_i}\} \subset \mathcal{V}$. At speed v, its $N_i(v)$ fitting transitions comprise measured states $y_{i,0}, \dots, y_{i,N_i(v)}$ and applied inputs $u_{i,0}, \dots, u_{i,N_i(v)-1}$. Section IV specifies their acquisition.

We identify a two-state affine model by output error (OE), comparing its simulated trajectory with the measurements. For one vehicle–speed pair, suppress i and v and write

$$\widehat{x}_{k+1} = A\widehat{x}_k + Bu_k + c, \qquad \widehat{x}_0 = \zeta, \tag{1}$$

where $A \in \mathbb{R}^{2\times 2}$, $B \in \mathbb{R}^{2\times 1}$, $c \in \mathbb{R}^2$, and $\zeta \in \mathbb{R}^2$ are fitted jointly. Define $W = \operatorname{diag}(\sigma_\beta^{-1}, \sigma_r^{-1})$, where $\sigma_\beta, \sigma_r > 0$ are the specified state measurement noise standard deviations. With $N = N_i(v)$, the fitting objective is

$$\mathcal{L}(A, B, c, \zeta) = \frac{1}{2(N+1)} \sum_{k=0}^{N} \|W(\widehat{x}_k - y_k)\|_2^2, \tag{2}$$

where $\|\cdot\|_2$ is the Euclidean norm and $\widehat{x}_k$ follows (1) without resetting to measured states. Minimizing (2) supplies $\widehat{A}_i(v), \widehat{B}_i(v)$ using the multistart procedure in Section IV. The offset c avoids requiring a known equilibrium center. Although each maneuver starts at zero lateral state, ζ must describe the fitting window's initial state after settling.

Let $A_i(v) \in \mathbb{R}^{2\times 2}$ and $B_i(v) \in \mathbb{R}^{2\times 1}$ denote the reference local perturbation matrices about the prescribed cornering condition. Collect their entries in $m_i(v) \in \mathbb{R}^{1\times 6}$ in the order $[A_{i,11}, A_{i,12}, A_{i,21}, A_{i,22}, B_{i,1}, B_{i,2}]$. The same ordering of the OE estimates gives $\widetilde{m}_i(v)$. These estimates can differ from $m_i(v)$ through measurement noise, finite-amplitude approximation, and numerical optimization. Only the matrix estimates

enter the scheduling dataset $\mathcal{D}_i = \{(v, \widetilde{m}_i(v)) : v \in \mathcal{V}_i\}$; the fitted offsets and initial states remain local.

B. Incomplete scheduling information and shared maps

Let $\phi(v) \in \mathbb{R}^L$ contain L linearly independent scheduling functions shared by all vehicles. Define $\Phi_i \in \mathbb{R}^{n_i \times L}$ and $Y_i \in \mathbb{R}^{n_i \times 6}$ with rows $\phi(v_{i,\ell})^\top$ and $\widetilde{m}_i(v_{i,\ell})$, respectively, for $\ell = 1, \dots, n_i$. The setting of interest has

$$\operatorname{rank} \Phi_i < L. \tag{3}$$

A nonzero $U_i \in \mathbb{R}^{L \times 6}$ can then satisfy $\Phi_i U_i = 0$: coefficient maps differing by U_i agree at observed speeds but can predict differently elsewhere. Additional samples at the same speeds cannot remove this scheduling ambiguity. Local dynamical excitation and scheduling coverage are therefore separate requirements.

For K shared maps, let $g_i \in \{1, \dots, K\}$ assign vehicle i to a group with coefficients $\Theta_{g_i} \in \mathbb{R}^{L \times 6}$. Write

$$m_i(v) = \phi(v)^\top \Theta_{g_i} + d_i(v), \tag{4}$$

where $d_i(v) \in \mathbb{R}^{1 \times 6}$ combines basis approximation error and individual departures from shared dynamics. Complementary speed supports can supply missing scheduling directions, but pooling introduces this recipient-dependent mismatch, separate from local estimation error in Y_i.

C. Identification objective and scope

Given $\mathcal{D}_i$, we seek K, assignments g_i, and coefficients Θ_g yielding $\widehat{m}_i(v) = \phi(v)^\top \widehat{\Theta}_{g_i}$. Section III learns these maps from observed targets; evaluation tests accuracy at locally unobserved speeds $\mathcal{V} \setminus \mathcal{V}_i$ and the mismatch introduced by pooling.

Clients exchange only the statistics and scores defined in Section III. Recipient-specific parameters, reference matrices, and family labels are excluded from fitting and grouping. Each fleet learns maps for its contributing vehicles, so generalization concerns unobserved speeds. "Full envelope" denotes the speed interval along the prescribed cornering curve, not an arbitrary nonlinear state/input domain.

III. Federated Compatible Regression

Using the local targets Y_i and designs Φ_i from Section II, we alternate fitting shared maps and assigning vehicles by agreement at observed speeds. Penalized residual error selects the group count. This chooses sharing partners from partial observations; group coverage is assessed afterward rather than enforced.

A. Additive fitting and conditional identifiability

For a fixed partition, let $\mathcal{I}_g = \{i : g_i = g\}$ be the set of vehicles in group g. Its map minimizes the regularized least-squares objective

$$\widehat{\Theta}_g = \arg \min_{\Theta \in \mathbb{R}^{L \times 6}} \left\{ \sum_{i \in \mathcal{I}_g} \|Y_i - \Phi_i \Theta\|_F^2 + \rho \|\Theta\|_F^2 \right\}, \tag{5}$$

where $\|\cdot\|_F$ is the Frobenius norm and $\rho = 10^{-8} + 10^{-12}$ combines ridge regularization with a numerical diagonal adjustment. Each speed–matrix target has equal weight; OE estimation covariances are not used. Client i computes $G_i = \Phi_i^\top \Phi_i \in \mathbb{R}^{L \times L}$ and $H_i = \Phi_i^\top Y_i \in \mathbb{R}^{L \times 6}$. The server sums these statistics within each group and solves

$$\left(\sum_{i \in \mathcal{I}_g} G_i + \rho I_L \right) \widehat{\Theta}_g = \sum_{i \in \mathcal{I}_g} H_i, \tag{6}$$

where I_L is the $L \times L$ identity matrix. For a fixed partition, this gives the pooled ridge solution in exact arithmetic without sharing Y_i. Positive regularization ensures a unique solution, but it does not establish that the data determine all scheduling directions. The following standard least-squares rank condition makes this distinction explicit.

Proposition 1: For a fixed nonempty group $\mathcal{I}_g$, let Φ_g be the vertical stack of its local designs Φ_i. The unregularized least-squares fit ($\rho = 0$) has a unique minimizer if and only if $\operatorname{rank} \Phi_g = L$, equivalently $\operatorname{rank}(\sum_{i \in \mathcal{I}_g} G_i) = L$. If, in addition, the targets satisfy $Y_i = \Phi_i \Theta_g$ for every group member, that minimizer equals Θ_g.

Proof: The aggregate Gram matrix is $\Phi_g^\top \Phi_g$, which is positive definite exactly when Φ_g has full column rank. Otherwise, a nonzero $U \in \mathbb{R}^{L \times 6}$ with $\Phi_g U = 0$ can be added to any minimizer without changing its predictions. For exact shared-map targets, Θ_g achieves zero residual and is therefore the unique minimizer in the full-rank case.

The condition shows how pooling can resolve the missing scheduling directions in Section II. In the present regularized fits, however, conditioning, OE error, and the mismatch $d_i(v)$ also affect prediction accuracy. Group selection must therefore be evaluated beyond rank alone.

B. Assignment and model selection

Assignment compares how well each candidate map describes a vehicle's observed dynamics. Scaling balances the six matrix entries. For entry $j \in \{1, \dots, 6\}$, let τ_j be its standard deviation across all observed target rows, using the population divisor $\sum_i n_i$. The server computes these values from local row counts, entry sums, and squared sums. Define $s_j = \max\{\tau_j, 0.05 \operatorname{median}(\tau_1, \dots, \tau_6)\}$ and $D = \operatorname{diag}(s_1, \dots, s_6)$. Unlike the state-noise weights W in OE, D describes target variation across the fleet. Its entries are checked to be positive and fixed during each search. Given the broadcast maps, client i returns

$$q_{ig} = \|(Y_i - \Phi_i \widehat{\Theta}_g) D^{-1}\|_F^2, \quad g = 1, \dots, K, \tag{7}$$

and is assigned to the lowest-scoring group, with ties choosing the lowest index. Fitting uses the unscaled objective (5); we do not claim that these alternating steps minimize one common objective.

For initialization, fit all vehicles as one group to obtain the Global map $\widehat{\Theta}_0$. Client i sends its six-entry mean residual

$$a_i = \frac{1}{n_i} \sum_{\ell=1}^{n_i} \left[\widetilde{m}_i(v_{i,\ell}) - \phi(v_{i,\ell})^\top \widehat{\Theta}_0 \right]. \tag{8}$$

Algorithm 1: Federated Group Learning from Local OE Models

Input: Local designs Φ_i and OE matrix targets Y_i, $i = 1, \dots, M$; common basis and regularization ρ.
Output: Selected K, assignments g_i, and maps $\widehat{\Theta}_g$, $g = 1, \dots, K$.
Prepare statistics and initialization summaries
Clients: send $G_i = \Phi_i^\top \Phi_i$, $H_i = \Phi_i^\top Y_i$, row counts, and target entry sums and squared sums
Server: fit Global $\widehat{\Theta}_0$ by (6) with all vehicles; compute D (Section III-B); broadcast both
Clients: send mean residuals a_i from (8)
Server: standardize the entries of a_i across clients
Search group counts and initial assignments
for $K = 1, \dots, 5$ **do**
  Use one start for $K = 1$ and 12 seeded starts otherwise
  for *each start* **do**
    Server: set $g_i = 1$ if $K = 1$; otherwise initialize by K-means on the standardized summaries
    for $t = 1, \dots, 30$ **do**
      Discard this start if any group has fewer than five vehicles
      Clients: send G_i, H_i and current group tags
      Server: fit each group by (6); broadcast all maps
      Clients: return scores q_{ig} from (7) for every group
      Server: assign each i to the lowest-scoring group, choosing the lowest index in a tie
      Break if no assignment changed
    Discard this start if any final group has fewer than five vehicles
    If the last iteration changed assignments, refit final groups and recompute scores without reassignment
    Server: compute BIC by (9); save this candidate
  Keep the lowest-BIC candidate for this K, if any
Select and distribute the result
Server: select the lowest-BIC candidate across K; send assignments g_i; clients use their selected broadcast maps

The server standardizes each entry across clients to zero mean and unit variance, uses unit scale for constant entries, and applies K-means. This scaling is separate from D. No complete local LPV map is needed.

Algorithm 1 specifies the search. Define the residual sum $\mathrm{SSE} = \sum_{i=1}^{M} q_{i,g_i}$ and scalar target count $n = 6\sum_i n_i$. The Bayesian information criterion (BIC) based score is

$$\mathrm{BIC} = n \log\big(\max\{\mathrm{SSE}/n, 10^{-12}\}\big) + p_K \log n, \quad (9)$$

where $p_K = 6LK + K - 1$ counts map coefficients plus a nominal mixture-weight term. Mixture weights are not estimated, and entries of an OE target need not be independent. We therefore use this as a heuristic complexity penalty. The lowest score selects the restart and K, including Global pooling at $K = 1$, without a guarantee of global optimality or physical-family recovery. Minimum group size does not ensure full rank, and deficient groups are not replaced by a fallback model.

C. Exchanged information

For $L = 7$, G_i has 28 distinct entries and H_i has 42. Initialization adds counts, target moments, and residual summaries; assignment uses K scores per client and broadcasts K maps of $6L$ entries. These messages are independent of trajectory length, although candidate counts and restarts determine total communication. Trajectories and targets stay local in the distributed emulator. Shared statistics can still reveal information; secure aggregation and differential privacy are not implemented.

TABLE I
VEHICLE PARAMETER CENTERS USED TO GENERATE THE FLEET. ALL CLIENTS HAVE $l_f = 1.2$ M AND $l_r = 1.6$ M.

Center	m [kg]	I_z [kg m^2]	C_f	C_r
Nominal	1500	2500	80000	80000
Heavy	1725	3000	84000	80000
Handling	1425	2250	72000	88000

Stiffnesses in N/rad.

IV. BENCHMARK AND EVALUATION

The benchmark compares selective sharing with local extrapolation and global pooling at unobserved speeds. Identical OE targets isolate the sharing comparison from local estimation. The protocol is fixed after three pilot fleets and tested on ten independent confirmation fleets. Secondary diagnostics examine sensitivity to the local prior and limitations from group coverage.

A. Vehicle fleet and local speed coverage

The single-track plant uses the sideslip–yaw state $x = [\beta, r]^\top$ and steering input u from Section II, with sideslip measured at the center of mass. At fixed longitudinal speed v, its nonlinear dynamics are

$$\dot{\beta} = (F_f + F_r)/(mv) - r, \quad (10)$$

$$\dot{r} = (l_f F_f - l_r F_r)/I_z, \quad (11)$$

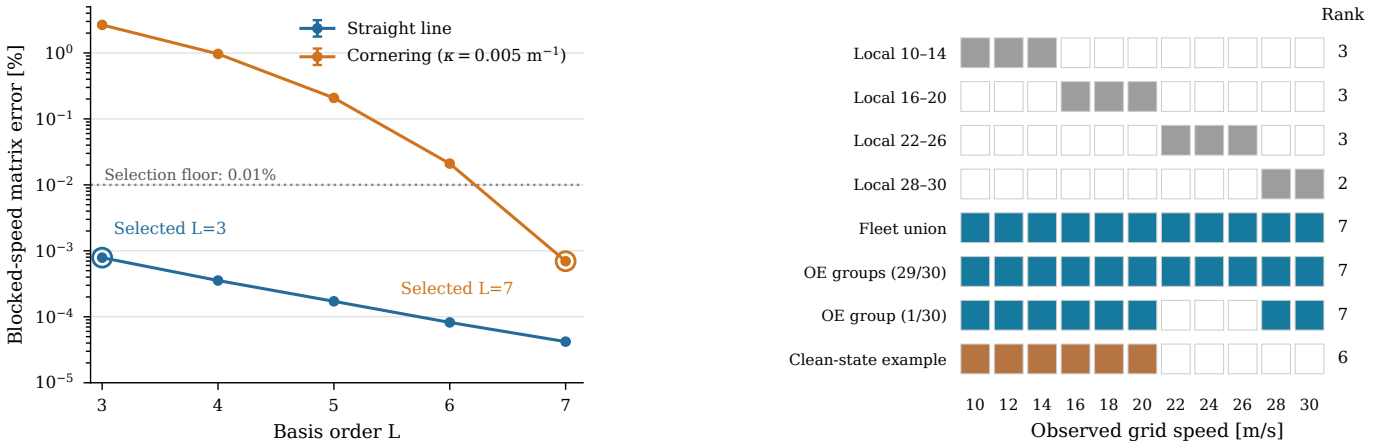
$$F_a = \mu F_{za} \tanh(C_a \alpha_a/(\mu F_{za})), \quad a \in \{f, r\}. \quad (12)$$

Here m is mass, I_z is yaw inertia, and l_f, l_r are the distances from the center of mass to the front/rear axles. For axle a, F_a, F_{za}, C_a, and α_a denote lateral force, normal load, cornering stiffness, and slip angle, respectively, with $\alpha_f = u - \beta - l_f r/v$, $\alpha_r = -\beta + l_r r/v$, $F_{zf} = mgl_r/(l_f + l_r)$, and $F_{zr} = mgl_f/(l_f + l_r)$. Gravity is $g = 9.81$ m/s^2 and the friction coefficient is $\mu = 0.9$. Angles β, u are in radians and r is in rad/s.

Each fleet contains 30 vehicles, with ten from each parameter center in Table I. Mass and inertia have independent uniform scatter of $\pm 2.5\%$, axle stiffnesses $\pm 5\%$, and geometry is fixed. Over $\mathcal{V} = [10, 30]$ m/s, the local blocks $\{10, 12, 14\}$, $\{16, 18, 20\}$, $\{22, 24, 26\}$, and $\{28, 30\}$ m/s are randomly assigned to eight, eight, seven, and seven clients. Family labels enter neither this allocation nor learned fitting. Each fleet therefore supplies 83 local vehicle–speed datasets.

B. Trajectory acquisition and local OE fitting

Each observed speed supplies a 44-s maneuver starting at $\beta = r = 0$. An LTI model of the Nominal vehicle (Table I), linearized at straight driving and 20 m/s, supplies one LQI gain and prefilter for all vehicles and speeds. The design weights (β, r, ξ) by $Q = \mathrm{diag}(25, 50, 1000)$ and steering by $R_u = 0.15$, where ξ integrates measured yaw-rate error. The integral is



(a) Development-only basis validation

(b) Scheduling information ($L = 7$)

Fig. 2. Scheduling representation and information coverage. (a) Earlier blocked-speed matrix error across ten development fleets; error bars denote one standard error. Rings mark the selected orders, and the dotted line is the 0.01% practical selection floor. (b) Filled cells mark observed grid speeds. The 30 primary OE groups have two coverage patterns, both with rank seven. The brown row is a secondary clean-state group in confirmation fleet 5 with rank six. Full rank does not remove within-group mismatch or guarantee extrapolation accuracy.

limited to ± 0.6 rad with conditional anti-windup; steering is limited to $\pm 12^\circ$ and 30°/s.

The yaw-rate reference ramps to $v\kappa_0$ over the first 4 s, with $\kappa_0 = 0.005$ m^{-1}. After 20 s for settling and a further 4-s hold, a 20-s fitting window tracks

$$r_{\mathrm{ref}}(t) = v\kappa_0 + \frac{a_{\mathrm{exc}}}{v} h(t), \qquad a_{\mathrm{exc}} = 0.3 \text{ m/s}^2, \tag{13}$$

where t is time within the fitting window. The unit-peak waveform h sums sinusoids at 0.2, 0.5, 0.9, and 1.4 Hz with seeded random phases and 2-s sine-squared tapers. Validation uses a separate maneuver with a 0.15–1.2 Hz chirp and the same taper. Feedback generates steering; sideslip is not prescribed independently.

At $T_s = 0.01$ s, each fitting window supplies $N_i(v) = 2000$ consecutive transitions: 2001 measured state vectors and 2000 applied steering inputs. Measurement noise is independent zero-mean Gaussian across samples and components, with standard deviations 0.01° for β and 0.03°/s for r. Feedback uses these noisy states; OE uses the same states and known applied steering, without input measurement noise. Fourth-order Runge–Kutta integrates the plant continuously within each maneuver.

OE uses trust-region least squares with analytic sensitivities and up to 300 function evaluations per start: affine OLS (initial spectral radius capped at 0.995), $A = 0.9I_2$, $A = 0.98I_2$, and the nominal model. The converged solution with the lowest training loss supplies the local matrix target, without a stability constraint. The separate validation maneuver assesses prediction and acquisition feasibility. The study assumes sideslip sensing and uses simulated trajectories.

C. LPV representation and comparison methods

The common seven-function basis is

$$b(v) = [1, v^{-1}, v^{-2}, z, z^2, z^3, z^4]^\top, \quad z = (v - 20)/10. \tag{14}$$

Stacking $b(v)^\top$ on 401 uniformly spaced speeds gives $B_{\mathrm{ref}} = QR$, with column-orthonormal $Q \in \mathbb{R}^{401\times 7}$ and invertible $R \in \mathbb{R}^{7\times 7}$. We use $\phi(v)^\top = b(v)^\top R^{-1}$ to improve conditioning without changing the span. Earlier development-only blocked-speed validation of reference matrices selected $L = 7$ for cornering with a one-standard-error rule and error floor 10^{-4}, and $[1, v^{-1}, v^{-2}]$ for straight driving (Fig. 2(a)).

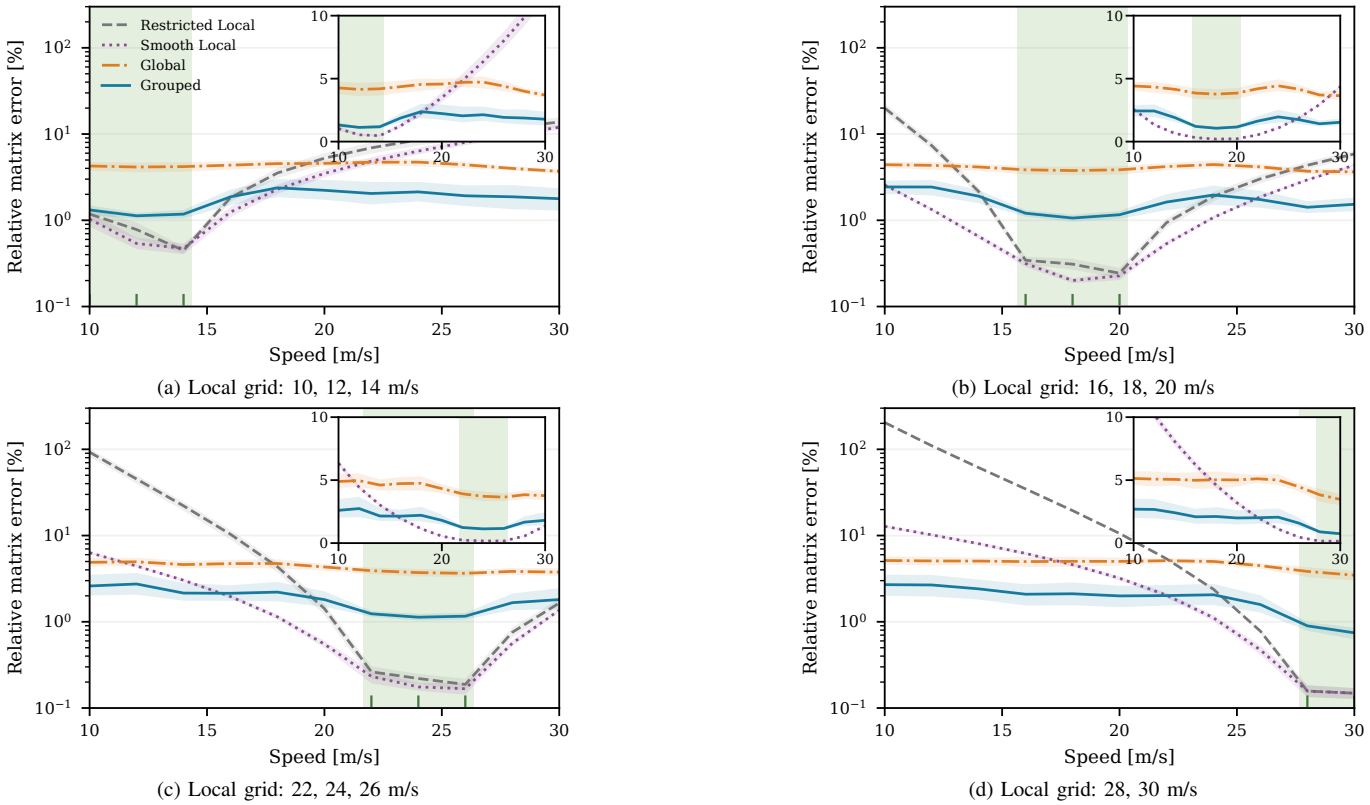
Restricted Local imposes a low-order prior, fitting the three straight-driving functions to each client's targets with ridge 10^{-8}. Global pools all targets with $L = 7$, while learned groups follow Section III. Smooth Local tests whether sharing adds value beyond a smoothness prior, using $L = 7$ and the curvature-penalized fit

$$\widehat{\Theta}_i^{\mathrm{S}} = \arg\min_{\Theta} \Big\{ \frac{\|Y_i - \Phi_i\Theta\|_F^2}{n_i} + \alpha\|Q_2\Theta\|_F^2 + \eta\|\Theta\|_F^2 \Big\}. \tag{15}$$

Here $Q_2 \in \mathbb{R}^{401\times 7}$ factors the trapezoidal approximation of $\frac{1}{2}\int_{-1}^{1} \|d^2[\phi(v(z))^\top\Theta]/dz^2\|_2^2\, dz$, where $v(z) = 20 + 10z$, and $\eta = 10^{-16}$ is numerical ridge. Affine functions in z remain unpenalized; the prior does not restore data rank. Earlier development selected α from $\{10^{-6}, 10^{-4}, 10^{-2}, 1, 10^2, 10^4, 10^6\}$ using unobserved-speed matrix error. The largest penalty within one standard error of the best fleet mean was 10^6, approaching an affine speed map. Confirmation uses this fixed penalty; all seven strengths form a secondary sensitivity check.

D. Evaluation metrics and statistical protocol

Reference matrix vectors $m_i^\star(v)$ come from numerical continuous-time Jacobians at $r_i^\star = v\kappa_0$, $\dot\beta = \dot r = 0$, discretized by an augmented matrix exponential at T_s. True parameters determine these evaluation references, not the acquisition controller or OE fits. On $\mathcal{V}_{\mathrm{grid}} = \{10, 12, \ldots, 30\}$ m/s, define



(a) Local grid: 10, 12, 14 m/s

(b) Local grid: 16, 18, 20 m/s

(c) Local grid: 22, 24, 26 m/s

(d) Local grid: 28, 30 m/s

Fig. 3. OE confirmation matrix error by local speed block. Light shading marks each observed block; short green marks identify its observed grid speeds. Curves average clients within block and fleet, then across ten fleets; bands are pointwise 95% fleet-bootstrap intervals from 20,000 draws, used descriptively. Insets show Global, Grouped, and Smooth Local on a linear 0–10% scale with identical averaging and bands. Smooth Local uses the frozen curvature penalty. These bands are distinct from the adjusted paired intervals for the primary endpoint.

the relative matrix error and primary unobserved-speed fleet error as

$$e_i(v) = \frac{\|\widehat{m}_i(v) - m_i^\star(v)\|_2}{\|m_i^\star(v)\|_2}, \qquad E_s^{\mathrm{U}} = \frac{1}{M}\sum_{i=1}^{M}\frac{1}{|\mathcal{U}_i|}\sum_{v\in\mathcal{U}_i} e_i(v), \tag{16}$$

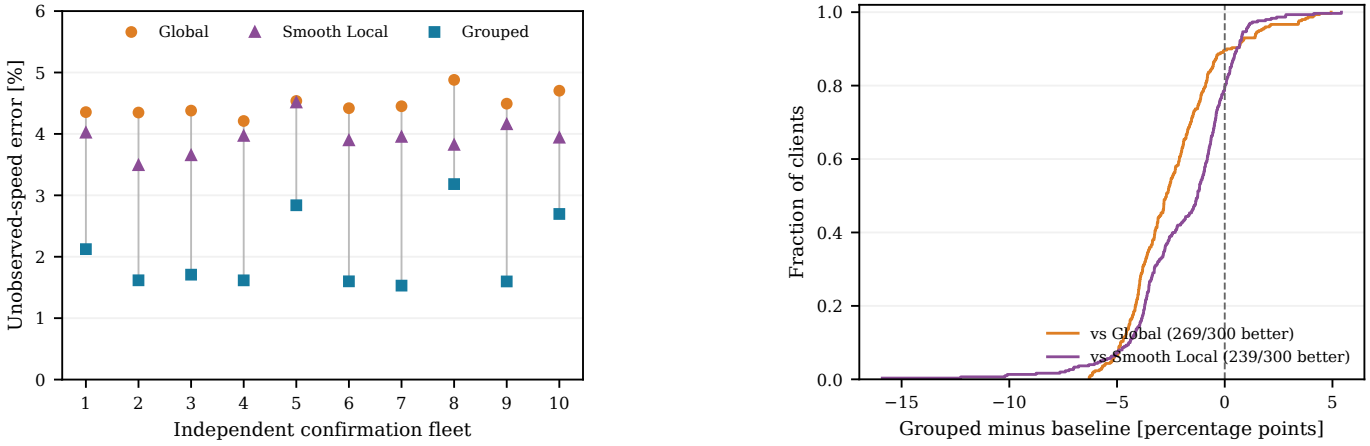
where $\mathcal{U}_i = \mathcal{V}_{\mathrm{grid}} \setminus \mathcal{V}_i$ and s indexes fleets ($M = 30$). Speeds are averaged within clients, clients equally within fleets, and fleets equally across the experiment. Observed-speed and full-grid errors replace $\mathcal{U}_i$ by $\mathcal{V}_i$ and $\mathcal{V}_{\mathrm{grid}}$, respectively. Separate relative A and B errors and entrywise absolute errors check whether the gains extend beyond the combined, state-unit-dependent metric. Tracking quality is evaluated separately in Section VI.

For each primary baseline $b \in \{\text{Smooth Local}, \text{Global}\}$, the paired difference is $\Delta_s^{(b)} = 100(E_s^{\mathrm{U},b} - E_s^{\mathrm{U,Fed}})$ in percentage points. Two Bonferroni-adjusted 97.5% two-sided Student-t intervals give simultaneous 95% family confidence under the paired-t assumptions, with fleets as sampling units. Confirmation requires both lower bounds above zero, ten valid fleet fits, and all equivalence checks passing. Inference with ten fleets is approximate. Percentage gains $100(1 - E_s^{\mathrm{U,Fed}}/E_s^{\mathrm{U},b})$ are averaged over fleets and reported descriptively.

E. Validity checks and secondary diagnostics

Acquisition validity is checked before comparing maps. Both training and validation maneuvers must satisfy steering limits, $|\beta| \le 5°$, $|a_y| \le 6$ m/s^2, and settling checks during the last 2 s before fitting, where $a_y = v(\dot{\beta} + r)$ is lateral acceleration. Settling requires mean yaw error at most $\max\{10^{-3}\,\mathrm{rad/s}, 0.01v\kappa_0\}$ and absolute fitted slopes of $\beta/(0.25°)$ and $r/(2°/\mathrm{s})$ at most $0.02\,\mathrm{s}^{-1}$. Actuator limiting is allowed because OE uses the applied input. A failed fit, bound, or settling check blocks that condition for the entire fleet. No cells or fleets are excluded or replaced, and no stability projection or Global fallback is used. Independent pooled augmented least squares must agree with Global and fixed-partition federated regression within 10^{-8} in coefficients and grid predictions.

Secondary diagnostics test whether accurate local targets or known vehicle classes suffice for extrapolation. Clean-state OLS uses noise-free simulator states and applied inputs from the same noisy-feedback trajectories, centered by preceding quiet-window means. Known-family pooling uses privileged labels. Coverage, rank, conditioning, and spectral radii diagnose failures on the same fleets without changing the primary method.



(a) Paired primary results

(b) Client tradeoff (descriptive)

Fig. 4. OE confirmation fleet gains and client tradeoffs. (a) Paired unobserved-speed errors; connectors join methods within each independent fleet. (b) Empirical cumulative distributions of client mean unobserved-speed error changes (Grouped minus baseline) in percentage points. Negative values indicate improvement. Client comparisons are descriptive; the 300 clients are nested within ten fleets.

V. Identification Results

A. Local fits and complementary coverage

All 830 local OE fits are valid, and all acquisition runs satisfy the applied-bound and settling checks. Local matrix error has median 0.247% and 95th percentile 1.206%. The steering limiter is active during 6.37% of validation samples on average, consistent with fitting the known applied input. The independent pooled solve for the same assignments agrees with the federated regression to within 5.12×10^{-14} in coefficients and 2.24×10^{-14} in grid predictions.

Accurate local matrices do not resolve missing speed information. For the selected seven-function basis, the local blocks have ranks three, three, three, and two, whereas their union has rank seven. Figure 2 relates the basis choice to this coverage gap.

B. Independent confirmation of sharing gains

Table II reports mean errors with the equal-client and equal-fleet weighting in Section IV. Grouped fitting reduces unobserved-speed error to 2.05%, compared with 3.94% for Smooth Local and 4.48% for Global. The paired absolute reductions are 1.89 percentage points against Smooth Local, with adjusted interval $[1.39, 2.39]$, and 2.43 points against Global, $[2.02, 2.83]$. Both lower bounds are positive, all ten fleets are valid, and all equivalence checks pass, satisfying the prespecified confirmation criterion. The corresponding mean paired percentage gains are 48.0% and 54.5%.

Local methods fit observed speeds more closely, but grouped sharing improves prediction outside those blocks (Fig. 3). Figure 4(a) shows improvement against both primary baselines in every fleet. Benefits are not uniform across recipients: 269/300 clients improve over Global and 239/300 over Smooth Local, with 31 and 61 worsening, respectively. Figure 4(b) shows these descriptive client-level changes: aggregate gains coexist with poorer representation of some recipients.

TABLE II
OE confirmation: mean relative matrix error [%] across ten independent fleets. Unobserved speed is the primary endpoint.

Method	Full grid	Observed	Unobserved
Restricted Local	16.02	0.38	20.44
Smooth Local	3.07	0.33	3.94
Global	4.34	3.87	4.48
Federated groups	1.82	1.09	2.05

C. What the learned groups capture

The selected groups supply the scheduling directions missing locally. Every confirmation fleet selects $K = 3$. All 30 group designs have rank seven, with condition numbers between 3.10 and 16.02 in the fixed QR basis. Twenty-nine groups contain all eleven grid speeds. The remaining group lacks observations at 22–26 m/s but has observations on both sides; its mean unobserved-speed error is 1.32%. Hence complementary observations can supply full scheduling rank without covering every grid point. This coverage emerged in the selected partitions; it was not enforced.

Family labels, inspected only afterward, give adjusted Rand indices from 0.518 to 1. Exact family recovery occurs in 4 of ten fleets. The other partitions mix physical families despite selecting three groups. The benefit therefore concerns sharing among vehicles selected from their observed dynamics, rather than reliable recovery of the generating vehicle classes.

D. Sensitivity to the local prior and error metric

Grouped fitting improves every fleet–penalty comparison across the seven prespecified Smooth Local penalties (70/70). The best aggregate confirmation penalty, 10^{-2}, gives 3.88% unseen error, compared with 3.94% for the frozen 10^{6} penalty. The mean paired gain against this retrospectively best setting remains 47.3%. This retrospective sensitivity check reuses the same fleets.

TABLE III

PRIMARY OE DIAGNOSTICS. A AND B COLUMNS ARE MEAN RELATIVE ERRORS [%] AT UNOBSERVED SPEEDS. THE LAST COLUMN COUNTS MATRICES WITH SPECTRAL RADIUS AT LEAST ONE OVER ALL 3300 CLIENT–GRID PAIRS.

Method	A error	B error	Unstable
Restricted Local	18.72	28.23	136
Smooth Local	2.73	10.98	272
Global	4.13	6.50	0
Federated groups	1.75	3.68	0

Errors fall for both A and B (Table III) and, in aggregate absolute terms, for each entry against both primary baselines. All grouped frozen-speed matrices also have spectral radius below one, with maximum 0.9765. Closed-loop consequences are tested in Section VI.

E. Coverage and conditioning limit extrapolation

Lower local error does not by itself ensure useful sharing. Clean-state OLS reduces median local matrix error to 0.052%, yet can yield deficient groups. In one fleet, a selected five-client group covers only 10–20 m/s and has rank six (Fig. 2(b)). Its unobserved-speed error is 38.76%, raising the fleet mean to 9.68%. Because the targets and assignments both change, this outcome illustrates a coverage failure without isolating the effect of measurement noise.

Known-family pooling is likewise not an accuracy bound. Two fleets have mean unseen errors of 44.19% and 23.35% when a physical family lacks observations at 10–14 m/s. These family designs remain full rank but have condition numbers above 3000. Physical similarity therefore need not supply a well-conditioned design over the intended interval. Local fit accuracy, group rank, and conditioning therefore require separate assessment.

VI. SUPPORTING LQI VALIDATION

We test yaw-rate tracking at locally unobserved speeds using the frozen maps and 300 confirmation vehicles, without retuning. Nominal LTI tests improvement over the acquisition controller; Global tests selective sharing.

A. Common controller design

The four identification methods and a privileged recipient-specific ExactLPV fit ($L = 7$) to the full reference grid supply scheduled controllers at 10, 12, …, 30 m/s. ExactLPV is a reference, not a performance bound. Nominal LTI keeps the 20-m/s acquisition gain and prefilter.

At each grid node, the fitted $\widehat{A}, \widehat{B}$ define

$$A_{\mathrm{a}} = \begin{bmatrix} \widehat{A} & 0 \\ T_s c_r & 1 \end{bmatrix}, \quad B_{\mathrm{a}} = \begin{bmatrix} \widehat{B} \\ 0 \end{bmatrix}, \quad c_r = [0, 1]. \tag{17}$$

The augmented state is $[\beta, r, \xi]^\top$, where ξ integrates yaw-rate error. Discrete Riccati synthesis uses the common acquisition weights from Section IV-B. For unit yaw reference, the equilibrium state $h \in \mathbb{R}^2$ and input $w \in \mathbb{R}$ solve $(I - \widehat{A})h - \widehat{B}w = 0$ and $c_r h = 1$. The prefilter is $F = w + K_x h$, where K_x contains the first two gain entries. Gains K and prefilters F are linearly interpolated between grid nodes.

For yaw reference r_k^{ref}, the runtime law is

$$\begin{aligned} u_k^{\circ} &= -K(v_k)[\beta_k, r_k, \xi_k]^\top + F(v_k) r_k^{\mathrm{ref}}, \\ \xi_{k+1}^{\mathrm{cand}} &= \mathrm{clip}_{[-0.6, 0.6]}\{\xi_k + T_s(r_k - r_k^{\mathrm{ref}})\}. \end{aligned} \tag{18}$$

Steering is rate-limited relative to u_{k-1}, then amplitude-limited. Conditional anti-windup accepts $\xi_{k+1}^{\mathrm{cand}}$ only when the command is unsaturated or the increment counteracts limiting; the bound is in radians. Feedback is noise-free. Only A, B enter control, not OE offsets or initial states.

TABLE IV

FIXED-SPEED TRACKING: MEAN YAW-RATE RMSE [DEG/S] OVER TEN REUSED FLEETS. PARENTHESES COUNT CONSTRAINT-VIOLATING RUNS OUT OF 300 PER SPEED. ALL FINITE RUNS CONTRIBUTE TO THE MEANS.

Method	13 m/s	21 m/s	29 m/s
Nominal LTI	0.0460 (0)	0.0142 (0)	0.0123 (0)
Restricted Local	11.7827 (56)	0.1133 (1)	1.3302 (18)
Smooth Local	0.0343 (0)	0.5294 (6)	5.6664 (68)
Global	0.0216 (0)	0.0140 (0)	0.0087 (0)
Federated groups	0.0207 (0)	0.0122 (0)	0.0068 (0)
ExactLPV reference	0.0204 (0)	0.0116 (0)	0.0064 (0)

B. Tracking at unobserved speeds

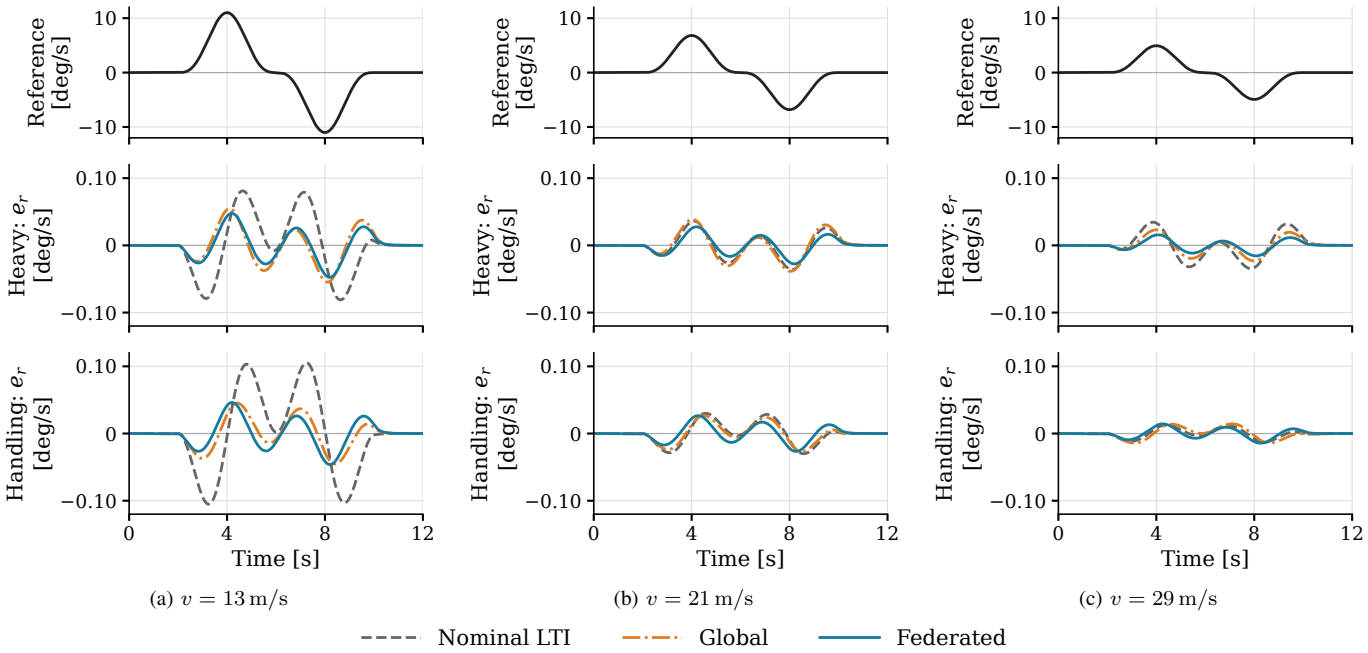
Separate 12-s runs at $v = 13, 21, 29$ m/s test interpolation between synthesis nodes at speeds unobserved locally by every vehicle. Each starts at $x_0 = 0$, $\xi_0 = 0$, $u_{-1} = 0$, with reference (time t in seconds)

$$r^{\mathrm{ref}}(t) = \frac{2.5}{v} \sin^3\left(\frac{\pi(t-2)}{4}\right), \qquad 2 \le t \le 10, \tag{19}$$

and zero otherwise, giving peak nominal lateral demand 2.5 m/s^2. RK4 uses $T_s = 0.01$ s with steering held within each step. Common bounds are 12° steering, 120°/s steering rate, and 6° sideslip; limiting and violations are recorded separately. Full-duration yaw RMSE includes finite infeasible responses, averaged first over clients and then fleets.

All 1800 schedules synthesize and all 5400 rollouts finish numerically. Nominal LTI, Global, Federated, and ExactLPV satisfy the bounds without limiting; their actual lateral acceleration and steering remain below 2.49 m/s^2 and 3.17°, respectively. Federated approaches ExactLPV and improves mean tracking over Nominal LTI at every speed (Table IV). Gains over Global are small in absolute terms, 0.00095–0.00194 deg/s, and occur in 9/10 fleets at 13 m/s and all ten at 21 and 29 m/s.

Figure 5 illustrates two recipients fixed before this evaluation. Heavy-family vehicles benefit most consistently; the handling-family mean slightly favors Global at 13 and 29 m/s. Restricted Local and Smooth Local violate bounds in 75/900 and 74/900 runs. Large means reflect a subset of control failures beyond credible vehicle operation. Successful synthesis alone does not ensure useful control outside local speed support.



(a) $v = 13\,\mathrm{m/s}$ (b) $v = 21\,\mathrm{m/s}$ (c) $v = 29\,\mathrm{m/s}$

Fig. 5. Fixed-speed yaw-rate tracking at three speeds unobserved in local identification. Top: reference r_{ref}. Middle and bottom: errors $e_r = r - r_{\mathrm{ref}}$ for the same fixed heavy and handling recipients, respectively. All six error panels use identical limits. The reference has the same nominal lateral-acceleration demand at each speed. These individual examples complement the fleet-level comparison.

C. Variable-speed stress tests and scope

Two supporting 12-s tests prescribe $11 \to 29 \to 15$ and $29 \to 12 \to 28$ m/s, with peak nominal lateral demand up to 7.7 m/s^2. All 3600 rollouts finish; Federated meets the bounds in all 600. Mean paired fleet-relative gains over Nominal LTI are 24.25% and 25.33%, versus 8.36% and 1.29% over Global. Specifically in these stress tests, all 100 heavy-family clients improve over Global and all 100 handling-family clients worsen, indicating maneuver-dependent recipient tradeoffs.

Prescribed-speed tests do not establish complete maneuver feasibility: longitudinal tire forces, braking load transfer, and combined slip are omitted. Nor do they establish nonlinear or varying-speed stability. Deployment needs recipient-level assessment, including estimation and path-tracking effects absent here.

VII. Discussion and Conclusion

Local output-error fitting and federated group regression combine complementary speed observations without exchanging trajectories, requiring complete local LPV models, or using physical family labels. Across ten independent simulated fleets, mean matrix error at locally unobserved speeds falls to 2.05%, versus 3.94% for Smooth Local and 4.48% for Global. Supporting LQI tests satisfy the tested constraints in all 1500 federated rollouts. Tracking improves over the fixed nominal controller; gains over Global are modest in absolute terms and depend on the recipient and maneuver.

Selective sharing thus improves full-envelope identification, but group selection guarantees neither adequate coverage nor better control for every recipient. All vehicles contribute to fitting, so transfer to new vehicles remains untested. Future work should assess measured vehicle data and observer-based feedback, and develop recipient-level checks for harmful transfer before deployment.

References

[1] R. Tóth, *Modeling and Identification of Linear Parameter-Varying Systems*, ser. Lecture Notes in Control and Information Sciences, vol. 403. Berlin, Germany: Springer, 2010, doi: 10.1007/978-3-642-13812-6.

[2] B. Bamieh and L. Giarré, "Identification of linear parameter varying models," *Int. J. Robust Nonlinear Control*, vol. 12, no. 9, pp. 841–853, 2002, doi: 10.1002/rnc.706.

[3] F. Ferranti and Y. Rolain, "A local identification method for linear parameter-varying systems based on interpolation of state-space matrices and least-squares approximation," *Mech. Syst. Signal Process.*, vol. 82, pp. 478–489, 2017, doi: 10.1016/j.ymssp.2016.05.037.

[4] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. AISTATS*, ser. Proc. Mach. Learn. Res., vol. 54, 2017, pp. 1273–1282.

[5] H. Wang, L. F. Toso, and J. Anderson, "FedSysID: A federated approach to sample-efficient system identification," in *Proc. L4DC*, ser. Proc. Mach. Learn. Res., vol. 211, 2023, pp. 1308–1320.

[6] A. Ghosh, J. Chung, D. Yin, and K. Ramchandran, "An efficient framework for clustered federated learning," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 19586–19597.

[7] L. F. Toso, H. Wang, and J. Anderson, "Learning personalized models with clustered system identification," in *Proc. IEEE CDC*, 2023, pp. 7162–7169, doi: 10.1109/CDC49753.2023.10383950.

[8] E. Keçeci, M. Güzelkaya, and T. Kumbasar, "Redefining clustered federated learning for system identification: The path of ClusterCraft," *arXiv preprint arXiv:2505.16857*, 2025.

[9] J. Weber et al., "Uncertainty-aware clustered federated identification for controller design: Application to lateral dynamics and yaw-rate tracking," accepted for publication in *Proc. IFAC World Congress*, 2026.